

Research Brief: federated learning for medical imaging

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Research question: what are the privacy risks?

Introduction

Federated learning (FL) has emerged as a promising approach to enable collaborative model training across distributed data sources while preserving privacy, particularly in sensitive domains like medical imaging. However, the inherent privacy risks in FL systems—especially when applied to healthcare data—remain inadequately understood and documented. This brief addresses the research question: *what are the privacy risks in federated learning for medical imaging* by synthesizing findings from recent literature on FL implementations and security vulnerabilities in medical contexts.

Summary

Both sources identify significant privacy risks in federated learning for medical imaging but frame them differently: [1] proposes VAFL as a method that mitigates these risks through perturbed local embedding and asynchronous operation, while [2] highlights that federated transfer learning (FTL) introduces unique security vulnerabilities due to domain shifts between clients. The literature shows that FL systems in medical imaging face risks from adversarial attacks and data heterogeneity, with [2] emphasizing that existing protocols are vulnerable to rough servers and adversarial parties, whereas [1] focuses on theoretical and empirical privacy guarantees for specific optimization objectives.

Findings

1. VAFL: a Method of Vertical Asynchronous Federated Learning

Tianyi Chen, Xiao Jin, Yuejiao Sun, Wotao Yin (2020)

Source reputation: medium — Open-access preprint server; dominant venue for ML/physics/CS research but not peer-reviewed

This paper introduces VAFL (Vertical Asynchronous Federated Learning), a method for vertical federated learning that operates asynchronously without client coordination. The method uses perturbed local embedding to ensure data privacy and improve communication efficiency while handling intermittent client connectivity. The approach provides theoretical convergence rates and privacy guarantees for strongly convex, nonconvex, and nonsmooth objectives, and empirically demonstrates competitive performance compared to centralized and synchronous FL methods on image and healthcare datasets.

Key claims:

- The new method allows each client to run stochastic gradient algorithms without coordination with other clients (p. 1)

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- The method uses a new technique of perturbed local embedding to ensure data privacy and improve communication efficiency (p. 1)

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- Theoretically, we present the convergence rate and privacy level of our method for strongly convex, nonconvex and even nonsmooth objectives separately (p. 1)

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- Empirically, we apply our method to FL on various image and healthcare datasets. The results compare favorably to centralized and synchronous FL methods (p. 1)

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2. Federated and Transfer Learning: A Survey on Adversaries and Defense Mechanisms

Ehsan Hallaji, Roozbeh Razavi-Far, Mehrdad Saif (2022)

Source reputation: medium — Open-access preprint server; dominant venue for ML/physics/CS research but not peer-reviewed

This survey examines security vulnerabilities and defense mechanisms in federated transfer learning (FTL), highlighting how FTL addresses data privacy constraints in federated learning by enabling knowledge transfer between clients with differing data attributes. The authors identify that FTL combines federated learning's data decentralization with transfer learning's ability to handle diverse data domains, but this integration creates unique security risks that require specialized defense strategies.

Key claims:

- The survey identifies that federated transfer learning (FTL) overcomes fundamental constraints of primary federated learning, particularly in terms of security. (p. 1)

This chapter performs a comprehensive survey on the intersection of federated and transfer learning from a security point of view. The main goal of this study is to uncover potential vulnerabilities and defense mechanisms that might compromise the privacy and performance of systems that use federated and transfer learning.

- FTL allows clients to differ in the employed attributes, which is more practical for industrial applications. (p. 2)

FTL clients may differ in the employed attributes, which is more practical for industrial application.

- FTL transmits information from the source domain's non-overlapping attributes to new samples within the target domain. (p. 4)

FTL takes a model that has been constructed on source data, and then aligns it to be employed in a target domain. This allows the model to be utilized for uncorrelated data points while exploiting the information gained from non-overlapping features in the source domain.

- Existing FL protocols are vulnerable to rough servers and adversarial parties that can compromise system functionality or privacy. (p. 5)

Existing FL protocol designs are vulnerable to rough servers and adversarial parties. While both infer confidential information from participants' updates, the former mainly tampers with the model training whereas the latter resorts to poisoning attacks to deviate the aggregation procedure.

Claims and hypotheses

- The new method allows each client to run stochastic gradient algorithms without coordination with other clients [1, p. 1]
- The method uses a new technique of perturbed local embedding to ensure data privacy and improve communication efficiency [1, p. 1]
- Theoretically, we present the convergence rate and privacy level of our method for strongly convex, nonconvex and even nonsmooth objectives separately [1, p. 1]
- Empirically, we apply our method to FL on various image and healthcare datasets. The results compare favorably to centralized and synchronous FL methods [1, p. 1]
- The survey identifies that federated transfer learning (FTL) overcomes fundamental constraints of primary federated learning, particularly in terms of security. [2, p. 1]
- FTL allows clients to differ in the employed attributes, which is more practical for industrial applications. [2, p. 2]
- FTL transmits information from the source domain's non-overlapping attributes to new samples within the target domain. [2, p. 4]
- Existing FL protocols are vulnerable to rough servers and adversarial parties that can compromise system functionality or privacy. [2, p. 5]

Conclusion

The gathered evidence supports that federated learning for medical imaging faces notable privacy risks from adversarial attacks and data heterogeneity, but the VAFL method provides a framework for asynchronous operation with privacy guarantees that could mitigate these risks under specific conditions.

Further research

- How do perturbed local embedding techniques in VAFL perform under real-world medical imaging data heterogeneity?
- What are the specific vulnerabilities of federated transfer learning systems when clients have non-overlapping data attributes?
- Can VAFL's theoretical guarantees be extended to handle adversarial attacks in medical imaging contexts?
- How do privacy risks in federated learning scale with the number of medical imaging

clients?

References

- Tianyi Chen, Xiao Jin, Yuejiao Sun, Wotao Yin (2020) 'VAFL: a Method of Vertical Asynchronous Federated Learning', arXiv:2007.06081v1. Available at: <http://arxiv.org/abs/2007.06081v1> [reputation: medium] [access: open]
- Ehsan Hallaji, Roozbeh Razavi-Far, Mehrdad Saif (2022) 'Federated and Transfer Learning: A Survey on Adversaries and Defense Mechanisms', arXiv:2207.02337v1. Available at: <http://arxiv.org/abs/2207.02337v1> [reputation: medium] [access: open]

Limitations

- **Published below the requested minimums** — 136/150 words, 2/2 reputable sources; minimums NOT met; search exhausted (no new sources, no new topics); loop limit reached (1/1) -> publishing
- *SupervisorAgent*: 136/150 words, 2/2 reputable sources; minimums NOT met -> researching further

Scope note: this is a deliberately small, local pipeline (free scholarly APIs + local retrieval + a local LLM). See the README for the design trade-offs.

Appendix: decision log

#	agent	kind	reason
1	ParseAgent	strategy	2007.06081v1: fallback to parse_basic (sweep)
2	ParseAgent	strategy	2207.02337v1: fallback to parse_basic (sweep)
3	SummariseAgent	summarised	The summary and key claims were extracted directly from the abstract section of the paper, which provides the primary research contributions and findings without additional context or interpretation.
4	SummariseAgent	summarised	The summary and claims were derived directly from the provided text, focusing on the survey's main purpose, FTL's characteristics, its operational mechanism, and identified vulnerabilities as explicitly stated in the text.
5	RankingAgent	ranked	2007.06081v1 (arxiv.org): medium — Open-access preprint server; dominant venue for ML/physics/CS research but not peer-reviewed [registry]
6	RankingAgent	ranked	2207.02337v1 (arxiv.org): medium — Open-access preprint server; dominant venue for ML/physics/CS research but not peer-reviewed [registry]
7	Indexer	indexed	2007.06081v1: 88 chunks into the vector store
8	Indexer	indexed	2207.02337v1: 95 chunks into the vector store
9	DiscoveryAgent	discovered	The keywords are specific terms, methods, and synonyms relevant to the paper's focus on vertical asynchronous federated learning, data privacy, and client connectivity challenges. Topics cover broader fields related to federated learning, distributed systems, and privacy preservation in machine learning contexts.